

Database System

Date:

Instructor(s)

Mr. Muhammad Naveed

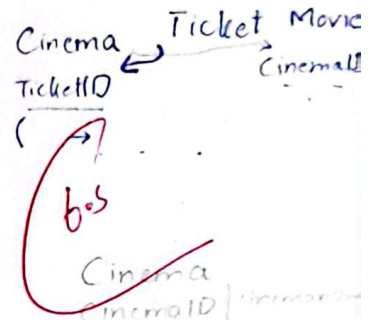
Quiz 01

Course code: CS2005

Total Time: 15 mins

Total marks: 10

Questions: 2



Roll No

Section

Student Signature

Attempt all the questions.

CLO5 Author queries using relational algebra and SQL

Q1: Consider the tables Cinema, Ticket, and Movie given below. ticketid in Cinema is a FOREIGN KEY referencing Ticket(ticketid) ON DELETE CASCADE, and cinemaID in Movie is a FOREIGN KEY referencing Cinema(cinemaID) ON DELETE CASCADE; foreign keys are written in *italics*, whereas primary keys are underlined.

Cinema (cinemaID, cinemaname, location, *ticketid*)

Ticket (ticketid, moviename)

Movie (movieid, moviename, *cinemaID*, ticketid)

a. Make ticketid in Movie a FOREIGN KEY referencing Ticket(ticketid) ON DELETE NO ACTION. (2 marks)

ALTER TABLE MOVIE

Add constraint t-m-fk

~~Foreign~~ FOREIGN KEY ticketid reference Ticket (ticketid)
ON DELETE NO ACTION

b. Explain what happens to tickets and movies if a cinema is deleted. (2 marks)

As when a cinema is deleted, its corresponding ticketID will also be deleted since its set on 'on delete cascade' so the corresponding ticketid in ticket table will also be deleted. Also since cinemaID for that corresponding cinema will also be deleted both in Cinema & Movie table

c. Explain what happens to movies if a ticket is deleted. (2 marks)

Movies would remain unchanged as there is no foreign key referencing ticket to Tickets table & so deleting a ticket won't affect movie as they're independent.

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Q2: From the following tables (USER and ArRATING), consider the query given below. Run the query and write the final output. (4 marks)

USER

UID	UName	Gender
1	Sara	F
2	Zara	F
5	Ali	M
3	Ahmad	M
9	Aliya	F
13	Tania	F
6	Hamza	M

ARTICLE (A)

ANo	Title	AuthorID	TNo
1	Why we need Relational Algebra?	9	3
2	All about Query Execution	13	8
3	What is Conceptual DB Model?	6	6
4	Nested SQL Queries	9	7
5	Transaction Processing	9	9

TOPIC (T)

TNo	TName	EditorID
1	Intro to DB	6
5	Basic SQL	5
7	Advanced SQL	9
9	Transactions	13
6	ER Model	1
8	Query Execution	13
3	Relational Algebra	9

ArRATING (R)

ANo	UID	Rating
1	2	3
1	9	3
3	2	3
2	2	4
2	6	5
2	13	2
3	13	5
4	2	5
4	6	5

⇒ 1, 1, 3, 2

SELECT R.ANo FROM ArRATING R WHERE R.Rating <= 3 UNION SELECT A.ANo FROM ARTICLE A WHERE A.TNo = (SELECT T.TNo FROM TOPIC T WHERE T.TName 'Transactions') ORDER BY ANo;

Output

write only once

1, 1, 2, 3, 9 → 5

(3)